

Md Mosharaf Hossan

Prescott, Arizona

208-406-7454 | mdmosharafhossan@isu.edu | mosharaf13.github.io

RESEARCH INTERESTS

Human-Computer Interaction, Virtual Reality, Empirical Software Engineering

EDUCATION

Master of Science in Computer Science | GPA: 3.96/4.0

Idaho State University

August, 2025

Idaho, United States

Bachelor of Science

Khulna University of Engineering & Technology

February, 2018

Khulna, Bangladesh

EMPLOYMENT

Assistant Professor

Embry-Riddle Aeronautical University

August 2025 – Present

Arizona, United States

Graduate Teaching Assistant

Idaho State University

August 2023 – May 2025

Idaho, United States

Mid-Senior Software Engineer

Nimble

May 2023 – July 2023

Bangkok, Thailand

Senior Software Engineer

Sigma Solutions

January 2021 – April 2023

Dhaka, Bangladesh

Software Engineer

Sigma Solutions

March 2019 – December 2020

Dhaka, Bangladesh

Software Engineer

Hyper Systems Ltd.

April 2018 – February 2019

Dhaka, Bangladesh

TEACHING EXPERIENCE

Assistant Professor

Embry-Riddle Aeronautical University

August 2025 – Present

Prescott, AZ

- **Courses, Fall 2025:** SIM251 Game Systems II Multi-player; SIM335 Game Engine Architecture; SIM351 Game Systems III Production
- **Courses, Spring 2026:** SIM150 Games Systems I; SIM205 Game Design Workshop; SIM421 Modeling & Simulation II; SIM450 Game Systems IV: Mixed Reality
- **SIM335 Game Engine Architecture**, where students apply design patterns, unit testing, DevOps workflows, AI coding tools, and Git/GitHub to build enterprise-grade game systems
- **SIM351 Game Systems III Production** and mentor students as they apply agile and Scrum practices to advance projects from prototype through development to release

- **SIM251 Game Systems II Multi-player**, where students to develop networked games with Unity and Photon while addressing persistence and synchronization challenges
- **SIM150 Games Systems I**, introducing foundational game systems concepts including player mechanics, state management, input handling, and basic data structures using Unity
- **SIM205 Game Design Workshop**, guiding students through iterative design, rapid prototyping, playtesting, and critique-driven refinement
- **SIM450 Game Systems IV: Mixed Reality**, where students build Virtual Reality applications using Unity and XR Interaction Toolkit with emphasis on spatial interaction and immersive system design
- **SIM421 Modeling & Simulation II**, focusing on full-stack development for interactive systems, including server-side programming with Laravel, RESTful APIs, databases, Unity client-server integration, and CI/CD
- Provide one-on-one mentorship during office hours to support technical and academic growth
- Review and grade assignments in Canvas, offer detailed feedback to strengthen student projects

Graduate Teaching Assistant

August 2023 – August 2025

Idaho State University

Pocatello, ID

- Delivered lectures for CS1181 Computer Science and Programming
- Served as a Product Owner for teams in CS3321: Introduction to Software Engineering course, guiding students in writing production-ready code using Unix, Git, GitHub, AI tools, unit testing, DevOps, and agile practices
- Held office hours to guide students in problem-solving
- Graded and provided constructive feedback on homework, assignments, and projects using Moodle learning management system
- Spring, 25: CS3321 Introduction to Software Engineering
- Fall, 24: CS4412/5512 Advanced Algorithms
- Spring, 24: CS4412/5512 Advanced Algorithms
- Fall, 23: CS1181 Computer Science and Programming

PEER REVIEWED PUBLICATIONS

- [COMPSAC, 25] Md Mosharaf Hossan, Rifat Ara Tasnim, Farjana Z. Eishita (2025). **Can You Keep Calm?: Adaptive Gameplay using Heart Rate as a Controller**. In *Proceedings of the 2025 IEEE 49th Annual Computers, Software, and Applications Conference (COMPSAC)*, pages 1-5, IEEE, 2025, Location. DOI:10.1109/COMPSAC65507.2025.00313
- [IETC, 25] Md Mosharaf Hossan, Rifat Ara Tasnim, Farjana Z. Eishita (2025). **Gamified Virtual Reality Exposure Therapy for Mysophobia: Evaluating the Efficacy of a Simulated Sneeze Intervention**. In *Proceedings of the 2025 Intermountain Engineering, Technology and Computing (IETC)*, pages 1-5, IEEE, 2025, Location. DOI:10.1109/IETC64455.2025.11039392
[[Premium Paper Award](#)]

- [IETC, 25] Aney Rani Paul, Md Mosharaf Hossan, Farjana Z. Eishita (2025). **Analyzing the User Experience of Google Drive Storage Management with Alert Notification**. In *Proceedings of the 2025 Intermountain Engineering, Technology and Computing (IETC)*, pages 1-5, IEEE, 2025, Location. DOI:10.1109/IETC64455.2025.11039452
- [MSR, 25] Costain Nachuma, Md Mosharaf Hossan, Asif Kamal Turzo, Minhaz F. Zibran (2025). **Decoding Dependency Risks: A Quantitative Study of Vulnerabilities in the Maven Ecosystem**. In *Proceedings of the 22nd international conference on mining software repositories (MSR)*, pages 270-274, IEEE, 2025, Location. DOI: 10.1109/MSR66628.2025.00048.
- [IoTaIS, 24] Md Mosharaf Hossan, Mostafa M. Fouda, Farjana Z. Eishita (2024). **Adaptive Game Design Using Machine Learning Techniques: A Survey**. In *Proceedings of The 2024 IEEE International Conference on Internet of Things and Intelligence System (IoTaIS)*, pages 144-150, IEEE, 2024, Location. DOI: 10.1109/IoTaIS64014.2024.10799330.
- [IETC, 24] Rifat Ara Tasnim, Md Mosharaf Hossan, Farjana Z. Eishita (2024). **Analyzing Differential Impact of Text-Based Instructions in Video Games**. In *Proceedings of the 2024 Intermountain Engineering, Technology and Computing (IETC)*, pages 227-232, IEEE, 2024, Location. DOI: 10.1109/IETC61393.2024.10564249. [\[Best Paper Award\]](#)

PROFESSIONAL EXPERIENCE

Researcher

August 2023 – August 2025

Idaho State University

Idaho, United States

- Designed and developed real-time VR and Android games that adapted to players' heart rate, enabling personalized exposure therapy for individuals with Mysophobia
- Mentored 3 undergraduate and graduate researchers, guiding them through VR-based research design, implementation, and data analysis for HCI studies
- Planned and conducted in-person between and within-subjects experiments to evaluate user responses to adaptive VR interventions

Mid-Senior Software Engineer

May 2023 – July 2023

Nimble

Bangkok, Thailand

- Initiated development of a google scrapping tool based on rails template.
- Contributed to Nimble's optimized rails template.

Senior Software Engineer

January 2021 – April 2023

Sigma Solutions

Dhaka, Bangladesh

- Led a 4-member Scrum team focused on elevating code quality and team synergy via rigorous code reviews
- Designed and implemented a concurrent diagnostic system for telecom nodes, integrating traffic, hardware, KPI, and alarm checks across legacy systems; reduced node diagnosis time from 1 hour to 5 minutes
- Built a near real-time monitoring bot for telecom outages, automatically detecting and correlating failures across network hierarchy, enabling faster response and saving around \$30,000 per month

- Automated telecom node creation workflows via browser automation and reduced manual workload by 90%
- Collaborated with cross-functional teams to design and deploy RESTful APIs for a SaaS customer query system, integrating 4 social platforms and improving agent response

Software Engineer

March 2019 – December 2020

Sigma Solutions

Dhaka, Bangladesh

- Launched a stock photo platform that scaled to 80K+ assets and grew into a leading regional marketplace
- Built a duplicate detection system for submitted photos using Rekognition and Elasticsearch, streamlining the image review workflow
- Delivered a CRM-driven operations platform that improved task scheduling and reduced coordination gaps
- Created and maintained internal packages to streamline development and enhanced team productivity

Software Engineer

April 2018 – February 2019

Hyper Systems Ltd

Dhaka, Bangladesh

- Developed RESTful backend APIs and frontend components for a real-time vehicle tracking system used by over 2,000 daily active devices
- Reduced false refuel alerts by 50% through sensor fluctuation analysis and delayed level validation

TECHNICAL SKILLS

Languages: Python, PHP, JavaScript, C/C++, C#, Java, SQL, L^AT_EX

Frameworks: Laravel, PHPUnit, Vue.js, Angular, Nuxt.js, D3.js, Tailwind CSS

DBMS: MySQL, PostgreSQL, MongoDB, Neo4j

Cloud: AWS(EC2, S3, Lambda, RDS)

Operating Systems: Linux, MacOS, Windows

Other Tools & Technologies:: Git, Docker, Docker Swarm, Unity, Redis, Elasticsearch, TDD, CI/CD, Shell Scripting, Moodle

HONORS & AWARDS

- Dean's Excellence Fund Award for attending COMPSAC 2025
- Premium Paper Award at IETC 2025
- Best Paper Award at IETC 2024

REVIEWING / COMMUNITY SERVICES

- COMPSAC 2025 Symposium on Mobile, Wearable & Ubiquitous Computing (MOWU)
- COMPSAC 2024 Symposium on Mobile, Wearable & Ubiquitous Computing (MOWU)
- 2024 IEEE Gaming, Entertainment, and Media Conference (GEM)
- 26th International Conference on Mobile Human-Computer Interaction (MobileHCI)

LEADERSHIP & VOLUNTEERING EXPERIENCE

- Treasurer, ISU Tennis Club — August 2023 – May 2025
- Volunteer, Bengal Bio Open House, Idaho State University (2025)
- Vice President, Bit to Byte, Software Development Club — August 2016 – July 2017
- Mentor, National High School Programming Contest (NHSPC) (2015, 2016, 2017).

REFERENCES

- **Dr. Farjana Z Eishita**
Assistant Professor, Department of Computer Science
Idaho State University
Email: farjanaeishita@isu.edu
- **Dr. Minhaz Zibran**
Associate Professor, Department of Computer Science
Idaho State University
Email: minhazzibran@isu.edu
- **Dr. Mostafa Fouda**
Associate Professor, Department of Computer Science
Idaho State University
Email: mfouda@isu.edu